SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Questions

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA05**

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images.* (BTL 3)

Assessment Tool Weightage: 50%

QUESTIONS: 5 **Total Marks50** **Duration :60 Minutes**

Annexure A

Scenario-Based Questions

Code of Conduct:

*I **Priyanga.B(192511102)**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Marks (100)**
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario	
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues	
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts	
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided	
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	

Scenario-Based Questions

1. Low Contrast Image

A grayscale image appears dull with very little difference between object and background intensities.

(a) **Clearly interpret the scenario and explain the cause of low contrast.**

(b) **Identify all key issues affecting image visibility.**

(c) **Apply an appropriate enhancement technique to address the problem.**

(d) **Justify your choice with logical reasoning.**

(e) **Present your explanation in a clear and structured manner.**

A grayscale image appears dull with very little difference between object and background intensities.

(a) Interpretation of scenario:

A low-contrast image occurs when the pixel intensity values are clustered within a narrow range of the total available scale (0–255) instead of spanning the full range. Visually, this appears dull because objects and background occupy nearly the same grey levels, so the eye cannot distinguish boundaries. The root cause is usually poor illumination during image acquisition, incorrect exposure/gain settings, a limited dynamic range sensor, or the image being formed largely of mid-grey tones without dark shadows or bright highlights.

(b) Key issues affecting visibility:

(i) Narrow histogram concentrated in a small intensity band

(ii) poor discriminability between object and background

(iii) loss of fine detail in shadow and highlight regions

(iv) reduced effectiveness of downstream operations such as edge detection, thresholding and segmentation, which rely on strong intensity differences.

(c) Applied enhancement technique:

Histogram Equalization (HE) is applied. HE computes the cumulative distribution function (CDF) of the input histogram and remaps each grey level so that the output histogram is spread as uniformly as possible across the full 0–255 range: $s = T(r) = (L-1) \times CDF(r)$, where r is the input intensity, s is the output intensity, and L is the number of grey levels. This stretches the clustered intensities apart, increasing global contrast.

(d) Justification:

Histogram Equalization is preferred over simple contrast stretching because it is fully automatic and does not require the user to specify minimum/maximum intensity limits; it adapts to the actual distribution of the image. It uses the entire dynamic range effectively and is computationally inexpensive, making it well suited for a dull, low-contrast grayscale image where the goal is a global, uniform improvement in contrast rather than enhancement of a specific local region.

(e) Structured summary:

Cause → narrow intensity range from poor illumination/exposure. Issue → indistinguishable object/background. Solution → Histogram Equalization redistributes intensities using the CDF. Result → improved global contrast and visibility

2. Uneven Illumination

An image captured under non-uniform lighting shows bright and dark patches.

(a) **Interpret the scenario and explain the impact of uneven illumination.**

(b) **Identify key factors causing intensity variation.**

(c) **Apply a suitable enhancement method to normalize illumination.**

(d) **Justify your selected approach with reasoning.**

(e) **Provide a clear and well-structured explanation.**

An image captured under non-uniform lighting shows bright and dark patches.

(a) Interpretation of scenario:

Under the image formation model, an image $f(x,y)$ can be expressed as the product of illumination $i(x,y)$ and reflectance $r(x,y)$: $f(x,y) = i(x,y) \times r(x,y)$. When the light source is not uniformly distributed (e.g., a single directional light, shadows, or lens vignetting), the illumination component $i(x,y)$ varies slowly across the scene, producing bright patches where lighting is strong and dark patches where it is weak, even though the underlying object reflectance is unchanged. This impacts visibility because the same object surface appears at different intensities depending on location.

(b) Key factors causing intensity variation:

(i) Non-uniform placement or angle of the light source

(ii) shadows cast by objects or the camera setup

(iii) vignetting/lens fall-off near image edges

(iv) sensor gain variations, all of which cause the slowly-varying illumination component to dominate over the true reflectance information.

(c) Applied enhancement method:

Homomorphic filtering is applied to normalize illumination. The method takes the logarithm of the image to convert the multiplicative illumination-reflectance model into an additive one: $\ln f = \ln i + \ln r$. A high-pass filter is then applied in the frequency domain to attenuate the slow-varying illumination component (low frequencies) while preserving/boosting the reflectance component (high frequencies), and finally the exponential is taken to return to the spatial domain.

(d) Justification:

Homomorphic filtering is chosen because it directly targets the physical cause of the problem — the multiplicative illumination component — rather than only adjusting the histogram globally. Since illumination varies slowly (low spatial frequency) and reflectance/object detail varies quickly (high spatial frequency), separating them in the frequency domain after the log transform allows illumination to be suppressed and fine detail to be enhanced simultaneously, producing a more uniformly lit image.

(e) Structured summary:

Cause → multiplicative illumination component varying spatially. Issue → bright/dark patches masking true reflectance. Solution → Homomorphic filtering (log → high-pass filter → exponential). Result → normalized, uniformly illuminated image.

3. Salt-and-Pepper Noise

A scanned document contains salt-and-pepper noise affecting readability.

(a) **Explain the scenario and characteristics of this noise.**

(b) **Identify key issues impacting image quality.**

(c) **Apply an appropriate filtering technique.**

(d) **Justify why this method is more suitable than alternatives.**

(e) **Present your response clearly and logically.**

A scanned document contains salt-and-pepper noise affecting readability.

(a) Scenario and noise characteristics:

Salt-and-pepper noise (impulse noise) appears as randomly scattered black (pepper) and white (salt) pixels superimposed on the image, typically caused by faulty scanner sensors, bit errors during transmission, or dust/dirt on the document during scanning. Unlike Gaussian noise, only a small fraction of pixels are affected, but those pixels take extreme intensity values (0 or 255), which sharply degrades text readability.

(b) Key issues impacting quality:

(i) Isolated extreme-valued pixels disrupt character shapes

(ii) OCR/readability is affected since noisy pixels resemble stray ink marks or missing dots

(iii) the noise is statistically independent of neighbouring pixels, making it visually distracting despite affecting only a small percentage of the image.

(c) Applied filtering technique:

A Median Filter is applied. For each pixel, a neighbourhood window (e.g., 3×3) is considered, the intensity values within the window are sorted, and the centre pixel is replaced with the median value of the neighbourhood. Since salt-and-pepper pixels are extreme outliers, they are effectively excluded from the median and removed.

(d) Justification over alternatives:

The median filter is more suitable than a mean (averaging) filter because the mean filter incorporates the extreme noisy values into its calculation, which spreads/blurs the noise instead of removing it and also blurs edges/text strokes. The median filter, being a non-linear, order-statistic-based operator, discards outliers entirely while preserving sharp edges — making it ideal for impulse noise on text documents where edge/stroke sharpness must be retained for readability.

(e) Structured summary:

Cause → scanner/transmission-induced impulse noise. Issue → isolated extreme pixels harming readability. Solution → Median filtering. Reasoning → removes outliers without blurring edges, unlike mean filtering.

4. Blurred Edges

An image processed for noise removal appears overly smooth, causing edges to blur.

(a) **Interpret the scenario and explain why edges are blurred.**

(b) **Identify key issues related to smoothing and edge loss.**

(c) **Apply an appropriate technique to restore edges.**

(d) **Justify your decision with clear reasoning.**

(e) **Structure your explanation clearly.**

An image processed for noise removal appears overly smooth, causing edges to blur.

(a) Interpretation — why edges are blurred:

Noise removal is commonly performed using smoothing filters such as the mean or Gaussian filter, which reduce noise by averaging each pixel with its neighbours. Edges in an image correspond to high-frequency components (sharp intensity transitions). Because averaging suppresses high-frequency content along with the noise, the very edges that carry important structural information are smoothed out, resulting in a blurred appearance.

(b) Key issues related to smoothing and edge loss:

(i) Loss of high-frequency edge information alongside noise

(ii) reduced sharpness of object boundaries

(iii) a fundamental trade-off between noise suppression (favours smoothing) and detail/edge preservation (favours minimal smoothing), which the initial processing failed to balance.

(c) Applied technique to restore edges:

Laplacian sharpening is applied. The Laplacian is a second-order derivative operator that highlights regions of rapid intensity change: $\nabla^2 f = [f(x+1,y)+f(x-1,y)+f(x,y+1)+f(x,y-1)] - 4f(x,y)$. The sharpened image is obtained by adding the Laplacian back to the original (blurred) image: $g(x,y) = f(x,y) - c \cdot \nabla^2 f(x,y)$, which re-emphasizes edges while retaining the noise-reduced background.

(d) Justification:

Laplacian sharpening is chosen because it specifically targets high-frequency edge content that was lost during smoothing, without reintroducing the impulse/random noise that was already removed, since noise typically has much lower magnitude second-derivative response after smoothing than true edges. This restores perceptual sharpness while keeping the benefit of the earlier noise-removal step.

(e) Structured summary:

Cause → smoothing filter removed high-frequency edge content along with noise. Issue → blurred boundaries. Solution → Laplacian sharpening added back to the image. Result → restored edge definition with noise still suppressed.

5. High-Frequency Noise
A satellite image contains fine-grained noise affecting feature visibility.
(a) **Interpret the scenario and describe high-frequency noise.**
(b) **Identify the key issues affecting image clarity.**
(c) **Apply an appropriate frequency domain filtering method.**
(d) **Justify the choice of filter with reasoning.**
(e) **Present your answer in a clear and organized manner.**

A satellite image contains fine-grained noise affecting feature visibility.

(a) Interpretation and description of high-frequency noise:

In the frequency domain, an image can be decomposed into low-frequency components (smooth/slowly varying regions such as terrain and large surfaces) and high-frequency components (rapid intensity changes such as edges, textures, and fine detail). Fine-grained noise in a satellite image corresponds to unwanted rapid, random pixel-to-pixel intensity fluctuations — i.e., energy concentrated at high spatial frequencies — typically introduced by sensor electronics or atmospheric interference, which visually appears as speckle/grain overlaying genuine terrain features.

(b) Key issues affecting clarity:

(i) High-frequency noise overlaps in the frequency spectrum with genuine fine features (roads, small structures), making them hard to separate

(ii) the grainy appearance reduces interpretability of the image for feature extraction and analysis

(iii) straightforward spatial filtering may blur genuine fine detail if not carefully controlled.

(c) Applied frequency-domain filtering method:

A Low-Pass Filter is applied in the frequency domain using the Fourier Transform: the image is transformed via the 2D DFT/FFT, a low-pass filter mask (Ideal, Butterworth, or Gaussian) is applied to attenuate high-frequency components beyond a chosen cutoff frequency D0 while retaining low frequencies, and the inverse DFT is taken to reconstruct the denoised spatial-domain image. A Butterworth Low-Pass Filter, H(u,v) = 1 / [1 + (D(u,v)/D0)^2n], is specifically recommended.

(d) Justification for filter choice:

The Butterworth Low-Pass Filter is justified over the Ideal Low-Pass Filter because the Ideal filter has a sharp cutoff that introduces ringing artifacts (Gibbs phenomenon) in the spatial domain, which would create false patterns in the satellite image. The Butterworth filter provides a smooth, gradual transition between the passed and attenuated frequencies (controlled by the order n), effectively suppressing high-frequency noise while minimizing ringing and better preserving genuine edges compared to the Ideal filter, and offering more control than a fixed Gaussian filter through the adjustable order parameter.

(e) Structured summary:

Cause → noise energy concentrated at high spatial frequencies. Issue → grainy appearance obscuring fine terrain features. Solution → Fourier Transform + Butterworth Low-Pass Filter + Inverse Fourier Transform. Reasoning → smooth cutoff avoids ringing artifacts of the ideal filter.